

Homework 2 Answers

BSTA 550

Non-textbook problems

- #1: (a) 1 does not hold because $0 \neq \frac{1}{216}$ (b) 2 does not hold because $P(B \cap C) \neq P(B)P(C)$
- NTB # 2: 0.3916

Textbook problems

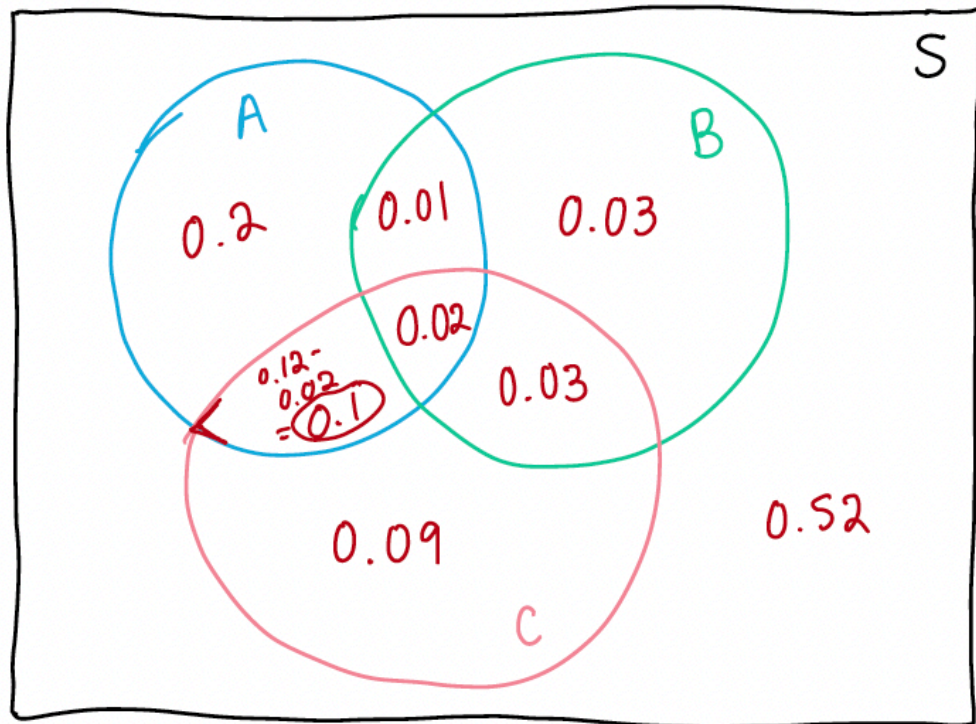
There are answers at the back of the book!! Selected answers (or hints) not provided at the end the book:

- Chapter 22
 - # 30: (a) 2,835 (b) 405 (c) 10,780 (d) 7,980
 - # 40: 0.6666667
 - # 42: 0.002116402 (This is the answer when $n = 5$. Your answer needs to be in terms of n .)
 - # 44: 0.3
 - # 46: 0.3333333
 - # 48: 0.007936508 (This is the answer when $n = 5$. Your answer needs to be in terms of n .)
- Chapter 3
 - # 4: (a) 0.111328 (b) 0.004872 0.995128
 - # 10: (c) 0.384 If you have the right answer to (c), then you should be able to figure out the rest (see (e)).

- # 12: No.
- NTB #1: (a) 0.0799 (b) 0.07553 (c) 0.0655
- Chapter 4
 - #4: 0.25
 - # 12: (a) 0.4285714 (b) 0.4285714 (c) 0.1428571
- Chapter 5
 - #17 a. 0.71; b. 0.29; c. 48/71; d. 1; e. 25/29; f. 25/39

From HW 1

Non-textbook problems



- # 1: (a)
- (b) Answers for part (ii) for each:
 1. 0.48
 2. 0.52

- 3. 0.2
- 4. 0.32
- 5. 0.03
- 6. 0.01
- 7. 0.02
- # 2: (a) 16, (c) for one, $P(X = 10) = \frac{5}{16}$, (d) 0.375

Textbook problems

There are some answers at the back of the book!! Selected answers (or hints) not provided at the end the book:

- Chapter 2
 - # 4: 0.35
 - # 8: 0.03125
 - # 16: 0.48
 - # 23: 0.47
 - # 30: (a) 0.189 (b) 0.811 (c) 0.189